Credit Scoring using Machine Learning

Proof of concept for a decision-making tool controlling risk

Yann HERENG
AI engineer training
OpenClassrooms
github.com/yhereng/OpenClassrooms

I. IDENTIFY BUSINESS PROBLEM AND POTENTIAL VALUE

Business use case

Bank Loans

Decision-making

YES

NO

Proposed solution

Supervised Classification

Data + ML

Added value

- Increase productivity
- Improve risk management
- Increase regulatory compliance

II. DATA PIPELINE

Multiple data sources

Merge / Aggregate

Missing value
One hot encoder
Label encoder
Feature engineering
Outliers
Scale
Standardize

III. METRIC SELECTION

Precision -Recall
ROC-AUC
Accuracy
Log-Loss

Fβ - score

Control over Precision and Recall

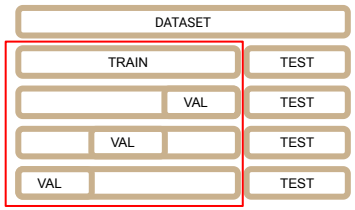
Trade off Coverage / Risk

VI. MODELS EVALUATION

Linear Models

LDA Classifier
SGD Classifier
Logistic Regression

Cross-validation



Best Model

Ensemble Methods

Random Forest Classifier
Light Gradient Boosting

II. OPTIMISATION

Hyperparameters

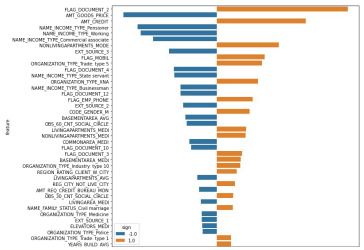
Random Search

Grid Search

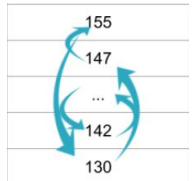
Best Hyperparameter set

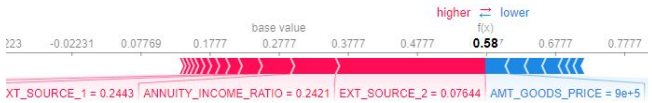
III. MODEL EXPLAINABILITY

Global level



Feature importance



Permutation importance

Local level